SAT Math Trick Pack 2

84 new challenge items · Answers and solutions

Name ..... Date .....

Each question states its tool: **No calculator needed**, **Calculator** or **Desmos**. Multiple-choice questions have one correct answer. For student-produced responses (SPR), write your answer in the box.

Algebra (20 items)

1. A-316 · Hard · SPR · **No calculator needed**

The function f is linear, $f(3) = k$ and $f(k) = 3$, where k is a constant and $k \neq 3$. If $f(10) = -1$, what is the value of k?

Answer: 6

The two points are $(3, k)$ and $(k, 3)$. Slope $= \dfrac{3-k}{k-3} = -1$ for every $k \neq 3$. So $f(x) = -x + b$. From $f(10) = -1$: $b = 9$, so $f(x) = -x + 9$ and $k = f(3) = 6$. Check: $f(6) = 3$ ✓.

Trap: Students think two unknowns (k and the slope) cannot be found, or assume the slope is $+1$ because the inputs and outputs are swapped.

---

2. A-317 · Hard · Multiple choice · **No calculator needed**

The function f is linear, $f(5) + f(-5) = 14$, and $f(2) = 13$. What is the value of $f(-2)$?

A) -13

B) 1

C) -1

D) 7

Answer: (B) 1

For $f(x) = mx + b$: $f(c) + f(-c) = 2b$ for every c. So $2b = 14$ and $f(2) + f(-2) = 14$, giving $f(-2) = 14 - 13 = 1$. ($m = 3$, $b = 7$.)

Trap: Students treat f as odd and answer -13, or try to find m and b from one point.

---

3. A-318 · Hard · SPR · **No calculator needed**

The function f is linear, $f(2) + f(8) = 30$, and $f(11) = 33$. What is the value of $f(0)$?

Answer: 0

For a linear function the average of two outputs is the output at the average input: $f(5) = \dfrac{30}{2} = 15$. Slope $= \dfrac{33-15}{11-5} = 3$, so $f(0) = 15 - 5 \cdot 3 = 0$.

Trap: Students think a sum of two outputs gives no usable point and stop, or use $(2, 30)$ as a point.

---

4. A-319 · Medium · Multiple choice · **No calculator needed**

In the xy-plane, what is the y-intercept of the graph of $x = 4y - 12$?

A) $(0, -12)$

B) $(0, -3)$

C) $(-12, 0)$

D) $(0, 3)$

Answer: (D) $(0, 3)$

Set $x = 0$: $4y = 12$, so $y = 3$. The y-intercept is $(0, 3)$. (Solving for y: $y = \frac{1}{4}x + 3$, so the slope is $\frac{1}{4}$, not 4.)

Trap: Students read the equation as if it were solved for y and take -12 as the y-intercept or 4 as the slope.

---

5. A-320 · Hard · Multiple choice · **No calculator needed**

In the equation $ax + by = c$, a, b and c are constants with $a > 0$, $b < 0$ and $c > 0$. The graph of the equation in the xy-plane does NOT pass through which quadrant?

A) Quadrant II

B) Quadrant III

C) Quadrant I

D) Quadrant IV

Answer: (A) Quadrant II

Solve for y: $y = -\frac{a}{b}x + \frac{c}{b}$. Since $b < 0$: slope $-\frac{a}{b} > 0$ and y-intercept $\frac{c}{b} < 0$. A line with positive slope and negative y-intercept passes through Quadrants I, III and IV and misses Quadrant II. Test: $a = 1, b = -1, c = 1$ gives $y = x - 1$.

Trap: Students forget that dividing by the negative b flips both signs, and read the slope as $-\frac{a}{b} < 0$.

---

6. A-321 · Hard · Multiple choice · **No calculator needed**

The system $2y - x > 4$ and $y < 5$ is graphed in the xy-plane. Which inequality describes all possible x-coordinates of points in the solution region?

A) $x > 6$

B) $x < 14$

C) $x < 6$

D) $x < -4$

Answer: (C) $x < 6$

From the first inequality, $x < 2y - 4$. Since $y < 5$, $2y - 4 < 6$, so every solution has $x < 6$. Every $x < 6$ is reached: for $x < 6$ choose y with $\frac{x + 4}{2} < y < 5$.

Trap: Students solve each inequality on its own (for example $x < -4$ from $y = 0$) instead of combining them through y.

---

7. A-322 · Hard · Multiple choice · **No calculator needed**

For real numbers x and y, $x + y < 12$ and $x - y > 4$. Which of the following must be true?

A) $y > 4$

B) $x > 8$

C) $y < 4$

D) $x < 8$

Answer: (C) $y < 4$

Rewrite the second inequality so both point the same way: $y - x < -4$. Adding $x + y < 12$ and $y - x < -4$ gives $2y < 8$, so $y < 4$. The value of x is not bounded: $(100, -90)$ satisfies both.

Trap: Students subtract the inequalities directly ($x + y - (x - y) < 12 - 4$) or add them as written to get a false bound on x.

---

8. A-323 · Hard · SPR · **No calculator needed**

If $-3 \le x \le 2$ and $-5 \le y \le 1$, what is the greatest possible value of xy?

Answer: 15

Check the corner products: $(-3)(-5) = 15$, $(-3)(1) = -3$, $2(-5) = -10$, $2(1) = 2$. The greatest is 15, from two negatives.

Trap: Students multiply the two upper bounds and answer 2.

---

9. A-324 · Medium · Multiple choice · **No calculator needed**

A farm has 40 hectares of date palms and 25 hectares of olive trees. The equation $40x + 25y = 3{,}600$ represents the farm's total harvest, in tonnes, last season. What does x represent?

A) The average date harvest per hectare, in tonnes

B) The number of tonnes harvested per date palm tree

C) The number of hectares of date palms

D) The total date harvest, in tonnes

Answer: (A) The average date harvest per hectare, in tonnes

40 is the number of date-palm hectares, so $40x$ is the date harvest in tonnes; x is the date harvest per hectare, in tonnes.

Trap: Because x is usually a count, students choose "the number of hectares", missing that the coefficients are the counts here.

---

10. A-325 · Medium · Multiple choice · **No calculator needed**

A line in the xy-plane has slope 3 and passes through the points $(2, k)$ and $(k, 14)$. What is the value of k?

A) 8

B) 4

C) 11

D) 5

Answer: (D) 5

$\frac{14-k}{k-2} = 3 \Rightarrow 14 - k = 3k - 6 \Rightarrow k = 5$. Check: points $(2, 5)$ and $(5, 14)$, slope $\frac{9}{3} = 3$ ✓.

Trap: Students write the slope as $\frac{k-2}{14-k}$ (run over rise) and get a wrong value such as 11.

---

11. A-326 · Hard · SPR · **No calculator needed**

In the xy-plane, the region with $x \geq 0$, $y \geq 0$ and $2x + y \leq 12$ is a triangle. The line $y = mx$ divides this triangle into two regions of equal area. What is the value of m?

Answer: 2

The vertices are $(0, 0)$, $(6, 0)$ and $(0, 12)$. A line through the vertex $(0, 0)$ halves the area exactly when it passes through the midpoint of the opposite side, $(3, 6)$. So $m = \frac{6}{3} = 2$. Check: the piece below the line, with vertices $(0, 0)$, $(6, 0)$ and $(3, 6)$, has area $\frac{1}{2} \cdot 6 \cdot 6 = 18$, half of 36 ✓.

Trap: Students halve a leg (for example the line through $(3, 0)$ or $(0, 6)$) instead of using the midpoint of the side opposite the origin.

---

12. A-327 · Medium · Multiple choice · **No calculator needed**

The function $V(t) = 1{,}200 - 0.75t$ gives the volume, in liters, of water in a tank t minutes after it starts draining. Which function gives the volume, in liters, h hours after the tank starts draining?

A) $1{,}200 - 0.75(h + 60)$

B) $1{,}200 - 45h$

C) $20 - 0.75h$

D) $1{,}200 - 0.0125h$

Answer: (B) $1{,}200 - 45h$

$t = 60h$, so $V = 1{,}200 - 0.75(60h) = 1{,}200 - 45h$.

Trap: Students divide the rate by 60 instead of multiplying, or shift the input.

---

13. A-328 · Hard · SPR · **No calculator needed**

In the system $y = 2x + k$ and $y = -x + 10$, k is a constant. For how many integer values of k is the solution (x, y) in Quadrant I (that is, $x > 0$ and $y > 0$)?

Answer: 29

$2x + k = -x + 10 \Rightarrow x = \dfrac{10 - k}{3}$ and $y = \dfrac{20 + k}{3}$. $x > 0 \Rightarrow k < 10$; $y > 0 \Rightarrow k > -20$. Integers from -19 to 9: 29 values.

Trap: Students include an endpoint ($k = 10$ or $k = -20$) or count $9 - (-19) = 28$.

---

14. A-329 · Medium · Multiple choice · **No calculator needed**

In the inequality $kx + 4 < 10$, k is a constant and $k < 0$. Which describes the solution set?

A) $x < \dfrac{6}{k}$

B) $x > \dfrac{6}{k}$

C) $x > -\dfrac{6}{k}$

D) $x < -\dfrac{6}{k}$

Answer: (B) $x > \dfrac{6}{k}$

$kx < 6$. Dividing by the negative k reverses the inequality: $x > \dfrac{6}{k}$. Test $k = -2$: $-2x < 6 \Rightarrow x > -3 = \dfrac{6}{-2}$ ✓.

Trap: Students divide by k without reversing the sign because the letter k looks positive.

---

15. A-330 · Medium · SPR · **No calculator needed**

A shopper spends exactly 600 dollars on x kilograms of oranges at 4 dollars per kilogram and y kilograms of apples at 5 dollars per kilogram, so $4x + 5y = 600$. If the shopper buys 10 more kilograms of oranges and still spends 600 dollars, by how many kilograms does y decrease?

Answer: 8

$4\Delta x + 5\Delta y = 0 \Rightarrow 4(10) + 5\Delta y = 0 \Rightarrow \Delta y = -8$: y decreases by 8 kg.

Trap: Students answer -10 (one-for-one trade) or -12.5 (dividing by 4 and multiplying by 5).

---

16. A-331 · Hard · Multiple choice · **No calculator needed**

If $7x - 2y = 3x + 6y$ and x and y are nonzero, what is the value of $\dfrac{x + y}{x - y}$?

A) The value cannot be determined from the information given.

B) $\dfrac{1}{3}$

C) 2

D) 3

Answer: (D) 3

$4x = 8y$, so $x = 2y$. Then $\dfrac{x + y}{x - y} = \dfrac{3y}{y} = 3$.

Trap: Students believe one equation in two unknowns cannot fix a value, and choose “cannot be determined”.

---

17. A-332 · Hard · Multiple choice · **No calculator needed**

Consider the system $3x - 4y = 10$ and $-6x + 8y = -20$. Which point lies on the graphs of both equations for every real number t?

A) $\left(t, \dfrac{3t - 10}{4}\right)$

B) $\left(t, \dfrac{10 - 3t}{4}\right)$

C) $\left(t, \dfrac{3t+10}{4}\right)$

D) $\left(\dfrac{4t-10}{3}, t\right)$

Answer: (A) $\left(t, \dfrac{3t-10}{4}\right)$

The second equation is -2 times the first, so the graphs are the same line and every point of $3x - 4y = 10$ works. Let $x = t$: $4y = 3t - 10$, so the point is $\left(t, \dfrac{3t-10}{4}\right)$. (With $y = t$, $x = \dfrac{4t+10}{3}$, not $\dfrac{4t-10}{3}$.)

Trap: Students make a sign error isolating y, or give up because the system has "no single solution".

---

18. **A-333** · Hard · Multiple choice · **No calculator needed**

A wall is $9x$ meters long, where x is a positive integer. Lamps are placed at both ends of the wall and every $\dfrac{9}{7}$ meters along it. Which expression gives the number of lamps?

A) $\dfrac{81x}{7} + 1$

B) $7x$

C) $7x + 1$

D) $9x + 1$

Answer: (C) $7x + 1$

Number of gaps $= \dfrac{9x}{9/7} = 7x$. With a lamp at both ends, lamps $=$ gaps $+1 = 7x + 1$. Check $x = 7$: 63 m, 49 gaps, 50 lamps.

Trap: Students count gaps instead of lamps ($7x$), or divide the wrong way ($\dfrac{81x}{7} + 1$).

---

19. **A-334** · Medium · Multiple choice · **No calculator needed**

How many solutions does the system $2x + y = 11$, $x - y = 1$, $x + 3y = -13$ have?

A) Exactly one

B) Exactly three

C) Zero

D) Exactly two

Answer: (C) Zero

The first two equations give $3x = 12$, so $(4, 3)$. Test it in the third: $4 + 3(3) = 13 \neq -13$. No point lies on all three lines: 0 solutions.

Trap: Students count the pairwise intersections (3) or stop after the first two equations (1).

---

20. **A-335** · Hard · SPR · **No calculator needed**

An oven's temperature is $70°$F when it is switched on at $t = 0$. For $3 \leq t \leq 12$, where t is the time in minutes, the temperature is modeled by $T = a + bt$, where a and b are constants. The model gives $T = 200$ at $t = 3$ and $T = 380$ at $t = 7$. What is the value of a?

Answer: 65

$b = \dfrac{380 - 200}{7 - 3} = 45$. $a = 200 - 45(3) = 65$. The model is valid only for $t \geq 3$, so its intercept 65 need not equal the real starting temperature 70.

Trap: Students assume the intercept must be the given starting value 70.

Advanced Math (19 items)

21. **AM-303** · Hard · Multiple choice · **No calculator needed**

In the equation $(k-2)x^2+4x+1=0$, k is a constant. The equation has exactly one real solution for exactly two values of k. What is the sum of these two values of k?

A) 8

B) 4

C) 2

D) 6

Answer: (A) 8

Case 1: $k=2$ makes the equation linear, $4x+1=0$, one solution. Case 2: $k\neq 2$ and $\Delta=16-4(k-2)=0$, so $k=6$. Sum $=2+6=8$.

Trap: Students use only the discriminant and miss that a zero leading coefficient makes the equation linear.

---

22. **AM-304** · Hard · SPR · **No calculator needed**

The function g is defined by $g(x)=\dfrac{x^2-kx-18}{x+3}$, where k is a constant. The graph of $y=g(x)$ in the xy-plane is a line with exactly one point removed. What is the y-coordinate of the removed point?

Answer: -9

The graph is a line only if $x+3$ cancels, so $x=-3$ is a zero of the numerator: $9+3k-18=0$, $k=3$. Then $x^2-3x-18=(x+3)(x-6)$ and $g(x)=x-6$ for $x\neq -3$. The removed point is $(-3,-9)$.

Trap: Students give the x-coordinate -3, or answer 0 because the function is undefined there.

---

23. **AM-305** · Hard · Multiple choice · **No calculator needed**

The function h is defined by $h(x)=\dfrac{x^2-4x-12}{x^2-36}$. What is the sum of all values of x for which $h(x)=0$?

A) 4

B) -8

C) 6

D) -2

Answer: (D) -2

The numerator is $(x-6)(x+2)$, but $x=6$ also makes the denominator 0, so $h(6)$ is undefined. The only zero is $x=-2$; the sum is -2.

Trap: Students set the numerator equal to 0 and use the sum of its roots, 4.

---

24. **AM-306** · Hard · Multiple choice · **No calculator needed**

The function h is defined by $h(x)=x^2-9$ for $x<2$ and $h(x)=-2x+10$ for $x\geq 2$. What is the sum of all values of m for which $h(m)=0$?

A) 0

B) 2

C) 5

D) 8

Answer: (B) 2

For $x<2$: $x^2=9$ gives $x=\pm 3$, but 3 is not less than 2, so only -3. For $x\geq 2$: $x=5$ ✓. Sum $=-3+5=2$.

Trap: Students keep $x=3$ although it is outside its piece, giving 5 or 8.

---

25. **AM-307** · Hard · Multiple choice · **No calculator needed**

In the equation $\frac{x^2}{x-a} = \frac{9}{x-a}$, a is a constant. For how many real values of a does the equation have exactly one real solution?

A) Infinitely many

B) Exactly two

C) Exactly one

D) Zero

Answer: (B) Exactly two

Multiplying gives $x^2 = 9$, so $x = \pm 3$, with the restriction $x \neq a$. The equation has exactly one solution when a equals one of these roots: $a = 3$ or $a = -3$. Two values.

Trap: Students treat the two roots as fixed and answer 0, or think every a works.

---

26. **AM-308** · Hard · Multiple choice · **No calculator needed**

The function f is defined by $f(x) = x^2 - 1$. How many distinct real solutions does the equation $f(f(x)) = 0$ have?

A) 1

B) 4

C) 2

D) 3

Answer: (D) 3

$f(u) = 0$ when $u = \pm 1$. $f(x) = 1 \Rightarrow x^2 = 2 \Rightarrow x = \pm\sqrt{2}$. $f(x) = -1 \Rightarrow x^2 = 0 \Rightarrow x = 0$ (one solution only). Total: 3.

Trap: Students expect $2 \times 2 = 4$ solutions, or stop at the zeros of f and answer 2.

---

27. **AM-309** · Hard · Multiple choice · **Desmos**

How many solutions (x, y) does the system $x^2 + y^2 = 25$ and $y = x^2 - 5$ have?

A) 3

B) 4

C) 1

D) 2

Answer: (A) 3

Substitute $x^2 = y + 5$: $y + 5 + y^2 = 25 \Rightarrow y^2 + y - 20 = 0 \Rightarrow y = 4$ or $y = -5$. $y = 4$: $x = \pm 3$ (two points). $y = -5$: $x^2 = 0$, $x = 0$ (one point). Total 3. Desmos: graph both and count the intersection points.

Trap: Students give every y-value two x-values (4) or count y-values only (2).

---

28. **AM-310** · Hard · Multiple choice · **Desmos**

How many real solutions does the equation $x^2 = 2^x$ have?

A) 1

B) 2

C) 3

D) 4

Answer: (C) 3

$x = 2$ and $x = 4$ work by inspection. For $x < 0$, x^2 grows to ∞ as $x \to -\infty$ while $2^x \to 0$, and at $x = 0$, $x^2 = 0 < 1 = 2^0$, so the graphs also cross once for $x < 0$ (near $x \approx -0.77$). Total 3. Desmos: graph $y = x^2$ and $y = 2^x$ and zoom out.

Trap: Students find 2 and 4 by testing integers and miss the negative root.

---

29. **AM-311** · Medium · SPR · **No calculator needed**

What is the sum of all solutions to the equation $(x-3)^2 = (2x+1)^2$?

Answer: $-\dfrac{10}{3}$ (or -3.333)

$x-3 = \pm(2x+1)$. With $+$: $x=-4$. With $-$: $x-3=-2x-1 \Rightarrow x=\frac{2}{3}$. Sum $= -4+\frac{2}{3} = -\frac{10}{3}$.

Trap: Students take square roots without $\pm$ and find only $x=-4$.

---

30. **AM-312** · Hard · Multiple choice · **No calculator needed**

How many real solutions does the equation $\sqrt{x^2-10x+25} = 5-x$ have?

A) Exactly two

B) Zero

C) Infinitely many

D) Exactly one

Answer: (C) Infinitely many

$\sqrt{x^2-10x+25} = \sqrt{(x-5)^2} = |x-5|$. And $|x-5| = 5-x$ exactly when $x-5 \le 0$, that is, for every $x \le 5$. Infinitely many solutions.

Trap: Students write $\sqrt{(x-5)^2} = x-5$ and get only $x=5$, or square both sides, reach $0=0$ and panic.

---

31. **AM-313** · Hard · Multiple choice · **No calculator needed**

A kiosk's daily profit, in hundreds of dollars, is modeled by $P(x) = -2x^2+12x+5$, where x is the price of an item in dollars. The model is used only for $4 \le x \le 7$. What is the greatest daily profit, in hundreds of dollars, predicted by the model on this interval?

A) 21

B) 23

C) -9

D) 5

Answer: (A) 21

The vertex is at $x = -\dfrac{12}{2(-2)} = 3$, outside the interval. On $4 \le x \le 7$ the parabola is decreasing, so the maximum is $P(4) = -32+48+5 = 21$.

Trap: Students use the vertex value $P(3)=23$ without checking the interval.

---

32. **AM-314** · Hard · Multiple choice · **No calculator needed**

The expression $\dfrac{6x^2+7x-5}{2x+3}$ is equivalent to $Ax+B+\dfrac{C}{2x+3}$, where A, B and C are constants. What is the value of $A+B+C$?

A) -4

B) 2

C) 4

D) 0

Answer: (D) 0

$(2x+3)(3x-1) = 6x^2+7x-3$, so $6x^2+7x-5 = (2x+3)(3x-1)-2$. $A=3$, $B=-1$, $C=-2$; sum $=0$.

Trap: Students use synthetic division with $-\frac{3}{2}$ and forget to divide the quotient by 2, getting $A=6$, $B=-2$.

---

33. **AM-315** · Hard · Multiple choice · **No calculator needed**

When the polynomial $p(x)$ is divided by $x - 5$, the remainder is 8. What is the remainder when $p(2x - 1)$ is divided by $x - 3$?

A) 15

B) 8

C) 5

D) 3

Answer: (B) 8

By the remainder theorem, the remainder is the value of $p(2x - 1)$ at $x = 3$, which is $p(5) = 8$.

Trap: Students transform the remainder ($2 \cdot 8 - 1 = 15$) or answer the input 5.

---

34. **AM-316** · Hard · Multiple choice · **No calculator needed**

For all real numbers x, $f(2x - 1) = 4x^2 + 3$. Which of the following defines $f(x)$?

A) $f(x) = x^2 + 4$

B) $f(x) = x^2 + 2x + 4$

C) $f(x) = 4x^2 + 3$

D) $f(x) = 4x^2 - 4x + 4$

Answer: (B) $f(x) = x^2 + 2x + 4$

Let $u = 2x - 1$, so $x = \dfrac{u + 1}{2}$ and $4x^2 = (u + 1)^2$. Then $f(u) = (u + 1)^2 + 3 = u^2 + 2u + 4$. Check: $x = 3$ gives $f(5) = 39$ and $25 + 10 + 4 = 39$ ✓.

Trap: Students copy the right side unchanged, or substitute $2x - 1$ into it instead of inverting.

---

35. **AM-317** · Hard · Multiple choice · **Desmos**

In the xy-plane, the graphs of $y = |x - 2|$ and $y = mx + 1$, where m is a constant, intersect at exactly two points. Which of the following could be the value of m?

A) -2

B) $-\dfrac{3}{4}$

C) 2

D) $\dfrac{1}{2}$

Answer: (D) $\dfrac{1}{2}$

Right branch ($x \geq 2$): $x = \dfrac{3}{1 - m} \geq 2$ needs $-\frac{1}{2} \leq m < 1$. Left branch ($x < 2$): $x = \dfrac{1}{1 + m} < 2$ needs $m > -\frac{1}{2}$ (or $m < -1$). Both branches meet the line when $-\frac{1}{2} < m < 1$, so $m = \frac{1}{2}$. Desmos: add a slider for m and count intersections.

Trap: Students think any line through $(0, 1)$ with a slope between -1 and 1 works, or check only one branch.

---

36. **AM-318** · Medium · SPR · **No calculator needed**

How many integers x satisfy the inequality $x(x - 4) < 2x$?

Answer: 5

$x^2 - 6x < 0 \Rightarrow x(x - 6) < 0 \Rightarrow 0 < x < 6$. Integers: $1, 2, 3, 4, 5$, so 5.

Trap: Students divide both sides by x (not allowed when x may be negative or zero) and get $x < 6$, which has infinitely many integers.

---

37. AM-319 · Hard · SPR · **No calculator needed**

The function f is defined by $f(x) = a\big((x-3)^2 - b\big)\big((x-3)^2 - c\big)$, where a, b and c are constants. If $f(1) = 40$ and $f(6) = -14$, what is the value of $f(5) + f(0)$?

Answer: 26

x appears only in $(x-3)^2$, so the graph is symmetric about $x = 3$. $f(5)$ uses $(5-3)^2 = 4$, the same as $f(1)$, so $f(5) = 40$. $f(0)$ uses 9, the same as $f(6)$, so $f(0) = -14$. Sum $= 26$.

Trap: Students try to solve for a, b and c (three unknowns, two equations) and decide it cannot be done.

---

38. AM-320 · Hard · SPR · **No calculator needed**

In the equation $\dfrac{x+a}{x^2-a^2} = -\dfrac{3}{20x}$, a is a constant. If $x = 6$ is a solution, what is the value of a?

Answer: 46

For $x \neq \pm a$, $\dfrac{x+a}{(x+a)(x-a)} = \dfrac{1}{x-a}$. So $\dfrac{1}{x-a} = -\dfrac{3}{20x} \Rightarrow 20x = -3x + 3a \Rightarrow a = \dfrac{23x}{3} = 46$. Check: $\dfrac{52}{36-2116} = -\dfrac{1}{40} = -\dfrac{3}{120}$ ✓.

Trap: Students cross-multiply the full quadratic and lose signs, or stop at a value of x.

---

39. AM-321 · Hard · Multiple choice · **No calculator needed**

In the equation $\dfrac{x^2}{\sqrt{x^2-k^2}} = \dfrac{k^2}{\sqrt{x^2-k^2}} + 24$, k is a positive constant. Which of the following is a solution?

A) $-\sqrt{k^2+576}$

B) $\sqrt{576-k^2}$

C) $\sqrt{k^2+24}$

D) $k+24$

Answer: (A) $-\sqrt{k^2+576}$

$\dfrac{x^2-k^2}{\sqrt{x^2-k^2}} = 24 \Rightarrow \sqrt{x^2-k^2} = 24 \Rightarrow x^2 = k^2 + 576 \Rightarrow x = \pm\sqrt{k^2+576}$. Only the negative root is offered. Check $k = 7$: $x = -25$.

Trap: Students forget to square 24 (choosing $\sqrt{k^2+24}$) or reject a negative answer.

Geometry and Trigonometry (26 items)

40. GT-232 · Hard · Multiple choice · **No calculator needed**

In the xy-plane, a circle has equation $(x-4)^2 + (y+2)^2 = 50$. Which point lies in the interior of the circle?

A) $(10, 2)$

B) $(9, 3)$

C) $(-1, -6)$

D) $(-3, 0)$

Answer: (C) $(-1, -6)$

Compare $(x-4)^2 + (y+2)^2$ with 50: $(-3, 0) \to 53$; $(9, 3) \to 50$ (on the circle); $(10, 2) \to 52$; $(-1, -6) \to 25 + 16 = 41 < 50$, inside.

Trap: Students check each coordinate separately ($|x-4| < \sqrt{50}$ and $|y+2| < \sqrt{50}$), choose a point that is on the circle, or flip the centre's signs.

---

41. GT-233 · Hard · SPR · **No calculator needed**

The graph of $x^2 + y^2 + 10x - 8y + k = 0$ in the xy-plane is a circle, where k is a constant. What is the greatest possible integer value of k?

Answer: 40

Complete the squares: $(x+5)^2 + (y-4)^2 = 41 - k$. A circle needs $r^2 > 0$, so $k < 41$. Greatest integer: 40. ($k = 41$ gives a single point.)

Trap: Students allow $r^2 = 0$ and answer 41, or reverse the inequality.

---

42. GT-234 · Hard · Multiple choice · **No calculator needed**

For $0 < t < \dfrac{\pi}{2}$, $\sin t = 0.6$. What is the value of $\cos(\pi + t) + \sin(\pi - t)$?

A) 0.2

B) -1.4

C) -0.2

D) 1.4

Answer: (C) -0.2

$\cos t = 0.8$. On the unit circle, $\pi + t$ is in Quadrant III, so $\cos(\pi + t) = -\cos t = -0.8$; $\pi - t$ is in Quadrant II, so $\sin(\pi - t) = \sin t = 0.6$. Sum $= -0.2$.

Trap: Students make both terms negative (-1.4) or both positive (1.4).

---

43. GT-235 · Hard · Multiple choice · **No calculator needed**

Points A, B and C lie on a circle with radius 12, and C lies on the major arc AB. The measure of angle ACB is $\dfrac{2\pi}{9}$ radians. What is the length of minor arc AB?

A) $\dfrac{16\pi}{3}$

B) $\dfrac{8\pi}{3}$

C) $\dfrac{64\pi}{3}$

D) $\dfrac{56\pi}{3}$

Answer: (A) $\dfrac{16\pi}{3}$

The central angle is twice the inscribed angle: $\dfrac{4\pi}{9}$. Arc length $= 12 \cdot \dfrac{4\pi}{9} = \dfrac{16\pi}{3}$.

Trap: Students use the inscribed angle as the central angle and get $\dfrac{8\pi}{3}$.

---

44. GT-236 · Hard · Multiple choice · **No calculator needed**

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that the measure of angle AED equals the measure of angle ABC. If $AD = 4$, $DB = 11$ and $AE = 6$, what is the length of $\overline{EC}$?

A) 10

B) 16.5

C) 22.5

D) 4

Answer: (D) 4

Triangles AED and ABC share angle A and have $\angle AED = \angle ABC$, so $A \leftrightarrow A$, $E \leftrightarrow B$, $D \leftrightarrow C$. Then $\dfrac{AE}{AB} = \dfrac{AD}{AC}$: $\dfrac{6}{15} = \dfrac{4}{AC}$, $AC = 10$, $EC = 10 - 6 = 4$. $\overline{DE}$ is NOT parallel to $\overline{BC}$.

Trap: Students assume $\overline{DE} \parallel \overline{BC}$ and use $\dfrac{AD}{DB} = \dfrac{AE}{EC}$, getting 16.5.

---

45. GT-237 · Hard · Multiple choice · **No calculator needed**

In right triangle ABC, angle C is the right angle, $AC = 16$ and $BC = 12$. Point E lies on $\overline{AC}$ with $AE = 10$, and point F lies on $\overline{AB}$ so that $\overline{EF} \perp \overline{AB}$. What is the length of $\overline{EF}$?

A) 8

B) 6

C) 7.5

D) 12.5

Answer: (B) 6

$AB = 20$. Triangles AFE and ACB are similar (shared angle A, right angles at F and C), with hypotenuses AE and AB. So $\dfrac{EF}{BC} = \dfrac{AE}{AB}$: $EF = 12 \cdot \dfrac{10}{20} = 6$.

Trap: Students match AE with the leg AC ($EF = 12 \cdot \frac{10}{16} = 7.5$) or give $AF = 8$.

---

46. GT-238 · Medium · Multiple choice · **No calculator needed**

A triangle has two sides of lengths 10 and 12, and the angle between them measures $150°$. What is the area of the triangle?

A) $30\sqrt{3}$

B) 30

C) 15

D) 60

Answer: (B) 30

The height to the side of length 10 is $12 \sin 150° = 12 \cdot \frac{1}{2} = 6$ (it falls outside the triangle). Area $= \frac{1}{2} \cdot 10 \cdot 6 = 30$.

Trap: Students use $\sin 60°$ or treat the triangle as right ($\frac{1}{2} \cdot 10 \cdot 12 = 60$).

---

47. GT-239 · Hard · SPR · **No calculator needed**

A rhombus has side length 13 and area 120. What is the sum of the lengths of its diagonals?

Answer: 34

Area $= \frac{1}{2} d_1 d_2 = 120 \Rightarrow d_1 d_2 = 240$. Half-diagonals and a side form a right triangle: $d_1^2 + d_2^2 = 4(13^2) = 676$. $(d_1 + d_2)^2 = 676 + 480 = 1156$, so $d_1 + d_2 = 34$ (diagonals 24 and 10).

Trap: Students forget the $\frac{1}{2}$ in the area formula or answer $2 \cdot 13 = 26$.

---

48. GT-240 · Hard · Multiple choice · **No calculator needed**

A right circular cone stands with its vertex pointing up. Its base radius is 6 and its height is 12. Water fills the cone to a depth of 6, measured from the base. What is the volume of the water?

A) 18π

B) 72π

C) 108π

D) 126π

Answer: (D) 126π

The empty top part is a cone similar to the whole with scale $\frac{1}{2}$, so its volume is $\frac{1}{8}$ of the cone. Cone volume $= \frac{1}{3}\pi(36)(12) = 144\pi$. Water $= \frac{7}{8} \cdot 144\pi = 126\pi$.

Trap: Students take the water as the small cone (18π) or halve the volume (72π).

---

49. GT-241 · Hard · SPR · **No calculator needed**

A sector of a circle has radius 10 and central angle $216°$. The sector is rolled into a right circular cone so that its two straight edges meet. The volume of the cone is $k\pi$. What is the value of k?

Answer: 96

Arc length $= 10 \cdot \dfrac{216}{360} \cdot 2\pi = 12\pi$, which is the base circumference, so $r = 6$. The sector radius is the slant height: $h = \sqrt{10^2 - 6^2} = 8$. $V = \frac{1}{3}\pi(36)(8) = 96\pi$.

Trap: Students use 10 as the cone's radius or height.

---

50. GT-242 · Hard · Multiple choice · **No calculator needed**

Line ℓ passes through $(0, 0)$ and $(a, 12)$, where $a > 0$. The acute angle between ℓ and the positive y-axis measures $60°$. What is the value of a?

A) $12\sqrt{3}$

B) $4\sqrt{3}$

C) 24

D) 6

Answer: (A) $12\sqrt{3}$

The angle with the positive x-axis is $90° - 60° = 30°$, so $\tan 30° = \dfrac{12}{a}$ and $a = \dfrac{12}{1/\sqrt{3}} = 12\sqrt{3}$.

Trap: Students use $60°$ as the angle with the x-axis and get $a = 4\sqrt{3}$.

---

51. GT-243 · Hard · Multiple choice · **No calculator needed**

In triangles PQR and XYZ, $PQ = XY = 9$ and $m\angle R = m\angle Z = 35°$, where $\overline{PQ}$ is opposite angle R and $\overline{XY}$ is opposite angle Z. Which additional fact is enough to prove that triangle PQR is congruent to triangle XYZ?

A) $PR = XZ$

B) $QR = YZ = 14$

C) $m\angle Q = m\angle Y$

D) $PR = QR$

Answer: (C) $m\angle Q = m\angle Y$

With $\angle Q = \angle Y$, the triangles have two angles and a non-included side equal (AAS). Choices with one more side give SSA, which can fit two different triangles (for $QR = 14$ there are two triangles, with $PR \approx 15.5$ or $PR \approx 7.4$). $PR = QR$ is about one triangle only.

Trap: Students accept side-side-angle as a congruence test.

---

52. GT-244 · Hard · Multiple choice · **No calculator needed**

A square has one side on the diameter of a semicircle of radius 10, and its other two vertices lie on the arc. What is the area of the square?

A) 100

B) 50

C) 80

D) 200

Answer: (C) 80

Let the side be s. A top vertex is $\frac{s}{2}$ from the centre horizontally and s up, so $s^2 + \left(\frac{s}{2}\right)^2 = 100$, $\frac{5}{4}s^2 = 100$, $s^2 = 80$.

Trap: Students use the full-circle result (area $2r^2 = 200$) or half of it.

---

53. GT-245 · Medium · Multiple choice · **No calculator needed**

In the xy-plane, the endpoints of a diameter of a circle are $(-2, 3)$ and $(6, -1)$. Which equation represents the circle?

A) $x^2 + y^2 - 4x - 2y - 15 = 0$

B) $x^2 + y^2 - 4x - 2y - 20 = 0$

C) $x^2 + y^2 - 4x - 2y - 75 = 0$

D) $x^2 + y^2 + 4x + 2y - 15 = 0$

Answer: (A) $x^2 + y^2 - 4x - 2y - 15 = 0$

Centre $(2, 1)$, $r^2 = (6-2)^2 + (-1-1)^2 = 20$. $(x-2)^2 + (y-1)^2 = 20 \Rightarrow x^2 + y^2 - 4x - 2y + 5 = 20 \Rightarrow x^2 + y^2 - 4x - 2y - 15 = 0$.

Trap: Students use the diameter squared (80) as r^2 or drop the constants $4 + 1$ when expanding.

---

54. GT-246 · Hard · Multiple choice · **No calculator needed**

Triangle ABC has area 60. Point D lies on $\overline{BC}$ with $BD : DC = 2 : 3$, and point E lies on $\overline{AD}$ with $AE : ED = 3 : 1$. What is the area of triangle BED?

A) 15

B) 18

C) 10

D) 6

Answer: (D) 6

Triangles ABD and ABC share the height from A: area $ABD = \frac{2}{5} \cdot 60 = 24$. Triangles BED and BAD share the height from B: area $BED = \frac{1}{4} \cdot 24 = 6$.

Trap: Students use the wrong part of a ratio ($\frac{3}{4}$ instead of $\frac{1}{4}$, or $\frac{2}{3}$ instead of $\frac{2}{5}$).

---

55. GT-247 · Hard · Multiple choice · **No calculator needed**

In a circle of radius 13, two parallel chords have lengths 24 and 10. Which of the following could be the distance between the two chords?

A) 12

B) 17

C) 26

D) 5

Answer: (B) 17

Distances from the centre: $\sqrt{169 - 144} = 5$ and $\sqrt{169 - 25} = 12$. Same side of the centre: $12 - 5 = 7$; opposite sides: $12 + 5 = 17$. Only 17 is offered.

Trap: Students picture only one case, or add the half-chord lengths.

---

56. GT-248 · Medium · Multiple choice · **Calculator**

The radius of a right circular cylinder is increased by 20% and its height is decreased by 25%. What is the effect on the volume of the cylinder?

A) It does not change.

B) It increases by 8%.

C) It decreases by 5%.

D) It increases by 19%.

Answer: (B) It increases by 8%.

V scales by $(1.2)^2(0.75) = 1.08$: an increase of 8%.

Trap: Students add percents $(20 - 25 = -5)$ or forget to square the radius factor.

---

57. GT-249 · Hard · Multiple choice · **No calculator needed**

Sphere Q has volume 36π. Sphere P has a radius 7 times the radius of sphere Q. The spheres are externally tangent at point T. Segment $\overline{MT}$ is a diameter of sphere P and segment $\overline{TN}$ is a diameter of sphere Q, with M, T and N on one line. What is the length of $\overline{MN}$?

A) 66

B) 45

C) 24

D) 48

Answer: (D) 48

$\frac{4}{3}\pi r^3 = 36\pi \Rightarrow r^3 = 27$, $r = 3$. Sphere P has radius 21. $MN = 2(21) + 2(3) = 48$.

Trap: Students add the radii (24) or solve $r^2 = 27$.

---

58. GT-250 · Hard · SPR · **Desmos**

In the equation $\sin\left(\frac{39\pi}{p}\right) = \frac{\sqrt{3}}{2}$, p is a constant and $p > 60$. What is the value of p?

Answer: 117

$p > 60$ means $0 < \frac{39\pi}{p} < \frac{39\pi}{60} = 0.65\pi$. The only angle in this interval with sine $\frac{\sqrt{3}}{2}$ is $\frac{\pi}{3}$ (the next, $\frac{2\pi}{3}$, gives $p = 58.5$). So $\frac{39}{p} = \frac{1}{3}$, $p = 117$.

Trap: Students take $\frac{2\pi}{3}$ and get 58.5, ignoring the condition, or work in degrees.

---

59. GT-251 · Hard · Multiple choice · **No calculator needed**

A rectangle is inscribed in a circle with area 676π. The length of the rectangle's diagonal is twice the length of its shorter side. What is the perimeter of the rectangle?

A) $52 + 52\sqrt{3}$

B) 104

C) $26 + 26\sqrt{3}$

D) $104 + 52\sqrt{3}$

Answer: (A) $52 + 52\sqrt{3}$

$r = 26$, so the diagonal (a diameter) is 52 and the short side is 26. The diagonal is twice the short side, so the triangle is $30°$–$60°$–$90°$ and the long side is $26\sqrt{3}$. Perimeter $= 52 + 52\sqrt{3}$.

Trap: Students assume a square, or use the radius as the diagonal.

---

60. GT-252 · Hard · SPR · **No calculator needed**

Two identical right rectangular prisms each have a square base with side length s and a height of 35. Each prism has surface area K. The prisms are glued together base to base, and the new solid has surface area $\frac{15}{8}K$. What is the value of s?

Answer: 10

$K = 2s^2 + 140s$. Gluing hides one base on each prism: new area $= 2K - 2s^2$. So $2K - 2s^2 = \frac{15}{8}K \Rightarrow \frac{1}{8}K = 2s^2 \Rightarrow K = 16s^2$. Then $2s^2 + 140s = 16s^2 \Rightarrow s = 10$.

Trap: Students subtract the glued face once, or set the new area to $2K$.

---

61. GT-253 · Hard · Multiple choice · **Calculator**

Square pyramids X and Y have the same base, a square with side length 10. Pyramid X has height 12 and pyramid Y has height 24. The total surface area of Y is approximately what percent greater than the total surface area of X?

A) 89%

B) 100%

C) 64%

D) 50%

Answer: (C) 64%

Slant heights: X: $\sqrt{12^2+5^2} = 13$; Y: $\sqrt{24^2+5^2} = \sqrt{601} \approx 24.52$. Areas: $X = 100 + 20(13) = 360$; $Y = 100 + 20(24.52) \approx 590.3$. $\dfrac{590.3}{360} \approx 1.64$: about 64% greater.

Trap: Students expect doubling (100%), use the height as the slant height, or compare lateral areas only (89%).

---

62. GT-254 · Hard · Multiple choice · **No calculator needed**

In triangle ABC, the measure of angle A is $18°$. Which additional fact is NOT sufficient to determine the measure of angle B?

A) $m\angle B = m\angle C$

B) $m\angle C = 3\,m\angle B$

C) Triangle ABC is isosceles.

D) $m\angle C = 90°$

Answer: (C) Triangle ABC is isosceles.

If the triangle is isosceles, angle A could be a base angle or the vertex angle, so B could be $18°$, $81°$ or $144°$. Each other fact fixes B: $B + 3B = 162 \Rightarrow B = 40.5°$; $B = C \Rightarrow B = 81°$; $C = 90° \Rightarrow B = 72°$.

Trap: Students think "isosceles" fixes the angles and stop at the first case.

---

63. GT-255 · Hard · Multiple choice · **Calculator**

Model Earth as a sphere with radius $6{,}371$ kilometers. An observer's eye is 1 kilometer above the surface. The line of sight from the eye to the horizon is tangent to Earth. Which is closest to the distance, in kilometers, from the eye to the point where the line of sight touches Earth?

A) 112.9

B) 1.0

C) $6{,}371.0$

D) 79.8

Answer: (A) 112.9

The radius to the point of tangency is perpendicular to the line of sight: $d = \sqrt{6{,}372^2 - 6{,}371^2} = \sqrt{12{,}743} \approx 112.9$.

Trap: Students compute $\sqrt{Rh} \approx 79.8$ (dropping the factor 2 in $2Rh$) or do not see the right triangle.

---

64. GT-256 · Hard · Multiple choice · **No calculator needed**

Parallel lines m and n are cut by a transversal. Angles b and c are same-side interior angles, and angle a is one of the other angles formed. If $200 < a + b + c < 236$, where all measures are in degrees, which of the following could be the measure of angle a, in degrees?

A) 56

B) 18

C) 64

D) 40

Answer: (D) 40

Same-side interior angles are supplementary: $b + c = 180$. So $200 < a + 180 < 236$, giving $20 < a < 56$. Only 40 works (56 is excluded by the strict inequality; 18 and 64 are outside).

Trap: Students include an endpoint or do not use $b + c = 180$.

---

65. GT-257 · Hard · Multiple choice · **No calculator needed**

In triangle JKL, the measure of angle J is $(ab)^\circ$ and the measure of angle K is $(cd)^\circ$, where $ab + cd = 90$. If $\sin J = 0.6$, what is the value of $\tan K$?

A) $\frac{3}{4}$

B) $\frac{4}{3}$

C) $\frac{4}{5}$

D) The value cannot be determined from the information given.

Answer: (B) $\frac{4}{3}$

$J + K = 90^\circ$, so angle $L = 90^\circ$ and J, K are complementary: $\cos K = \sin J = 0.6$, $\sin K = 0.8$, $\tan K = \frac{0.8}{0.6} = \frac{4}{3}$.

Trap: Students decide there is not enough information because a, b, c, d are unknown.

Problem-Solving and Data Analysis (19 items)

66. PS-157 · Hard · Multiple choice · **No calculator needed**

Data sets A and B each contain 20 values, all from the list $10, 20, 30, 40, 50$. The frequencies in A are $2, 3, 10, 3, 2$ and the frequencies in B are $5, 3, 4, 3, 5$, in that order. Which statement is true?

A) The standard deviations are equal because the means and ranges are equal.

B) The standard deviation of B is greater.

C) The standard deviation of A is greater.

D) The standard deviations cannot be compared without calculating them.

Answer: (B) The standard deviation of B is greater.

Both sets have mean 30 and range 40. B has more values at the extremes (10 and 50) and fewer at the mean, so its values are farther from the mean on average: SD of B (≈ 15.2) is greater than SD of A (≈ 10.5).

Trap: Students think equal means and ranges force equal standard deviations.

---

67. PS-158 · Medium · SPR · **No calculator needed**

The daily high temperatures in a city for one week have a mean of 34°C and a standard deviation of 4°C. Each temperature is converted to degrees Fahrenheit using $F = 1.8C + 32$. What is the standard deviation of the converted temperatures, in degrees Fahrenheit?

Answer: 7.2

Multiplying every value by 1.8 multiplies the SD by 1.8; adding 32 shifts the values but does not change the spread. SD $= 1.8 \cdot 4 = 7.2$.

Trap: Students apply the full conversion ($1.8 \cdot 4 + 32 = 39.2$) or think the SD does not change.

---

68. PS-159 · Hard · SPR · **No calculator needed**

The data set $12, 15, 19, 24, 30$ has a range of 18. A sixth value, x, is added, and the range of the new data set is 1.5 times the original range. What is the sum of all possible values of x?

Answer: 42

New range $= 27$. Either x is the new maximum: $x - 12 = 27$, $x = 39$; or the new minimum: $30 - x = 27$, $x = 3$. Sum $= 42$.

Trap: Students assume the new value is the maximum and answer 39.

---

69. PS-160 · Hard · SPR · **No calculator needed**

The data set $2, 6, 9, 15, x$ has a mean equal to its median, where x is a real number. What is the sum of all possible values of x?

Answer: 19

Mean $= \frac{32+x}{5}$. If $x \le 6$, median 6: $x = -2$ ✓. If $6 \le x \le 9$, median x: $32 + x = 5x$, $x = 8$ ✓. If $x \ge 9$, median 9: $x = 13$ ✓. Sum $= -2 + 8 + 13 = 19$.

Trap: Students assume x is the median, or reject the negative value.

---

70. PS-161 · Hard · SPR · **No calculator needed**

A data set contains the values $1, 2, 3, 4, 5$ with frequencies $4, k, 6, 3, 5$, respectively, where k is a positive integer. The median of the data set is 3. What is the greatest possible value of k?

Answer: 9

There are $18 + k$ values, and $4 + k$ of them are at most 2. For $k = 9$: 27 values, the 14th is 3 ✓. For $k = 10$: 28 values, the 14th is 2 and the 15th is 3, so the median is 2.5. Greatest $k = 9$.

Trap: Students ignore that the total changes with k, or forget that an even count averages the two middle values.

---

71. PS-162 · Hard · Multiple choice · **No calculator needed**

The commute times, in whole minutes, of 20 students are grouped: 10 students took 0 to 9 minutes, 7 took 10 to 19 minutes, and 3 took 20 to 29 minutes. Which of the following could NOT be the median commute time?

A) 10

B) 7.5

C) 14

D) 4

Answer: (D) 4

The median is the mean of the 10th and 11th values. The 10th is in 0–9 and the 11th is in 10–19, so the median is between $\frac{0+10}{2} = 5$ and $\frac{9+19}{2} = 14$. 4 is impossible; 7.5, 10 and 14 are possible.

Trap: Students think the median must lie inside one bin, or that it cannot be bounded at all.

---

72. PS-163 · Hard · Multiple choice · **No calculator needed**

Data set A has 11 values with median 40. Data set B has 11 values with median 52. The two data sets are combined. Which of the following could be the median of the combined 22 values?

A) 46

B) 38

C) 60

D) 54

Answer: (A) 46

At most 5 values of A and at most 5 values of B are less than 40, so at most 10 combined values are below 40 and the 11th value is at least 40. Likewise at most 10 combined values are greater than 52, so the 12th value is at most 52. The median lies between 40 and 52; only 46 is in this range.

Trap: Students think nothing can be said, or believe the median must equal the average of the two medians.

---

73. PS-164 · Hard · Multiple choice · **Calculator**

Electric vehicles made up 4% of a country's new-car sales in one year and 6% the next year. Total new-car sales increased by 10% from the first year to the second. By what percent did the number of electric vehicles sold increase?

A) 2%

B) 60%

C) 65%

D) 50%

Answer: (C) 65%

Let first-year sales be 100: EVs $= 4$. Second year: 110 cars, EVs $= 0.06(110) = 6.6$. Increase $= \dfrac{2.6}{4} = 65\%$.

Trap: Students answer 2 (percentage points) or 50 (ignoring the change in total sales).

---

74. PS-165 · Hard · SPR · **No calculator needed**

The number a is 150% more than the number b, and b is 60% less than the positive number c. The number a is what percent of c?

Answer: 100

$a = 2.5b$ and $b = 0.4c$, so $a = c$: 100%.

Trap: Students read "150% more than" as $1.5b$, giving 60%.

---

75. PS-166 · Hard · Multiple choice · **Calculator**

District A has $120{,}000$ residents and a population density of $4{,}000$ residents per square kilometer. District B has $60{,}000$ residents and a population density of $1{,}500$ residents per square kilometer. Which is closest to the population density, in residents per square kilometer, of the two districts combined?

A) $5{,}500$

B) $3{,}167$

C) $2{,}571$

D) $2{,}750$

Answer: (C) $2{,}571$

Areas: $\frac{120{,}000}{4{,}000} = 30$ and $\frac{60{,}000}{1{,}500} = 40$ km^2. Combined density $= \frac{180{,}000}{70} \approx 2{,}571$.

Trap: Students average the densities ($2{,}750$) or weight them by population ($\approx 3{,}167$).

---

76. PS-167 · Hard · SPR · **Calculator**

Bottle B holds 20% more juice than bottle A and costs 15% more. To the nearest tenth of a percent, by what percent is the price per liter of bottle B less than the price per liter of bottle A?

Answer: 4.2

Price per liter of B relative to A: $\dfrac{1.15}{1.20} \approx 0.9583$. It is about 4.2% less.

Trap: Students subtract the percents and answer 5.

---

77. PS-168 · Hard · SPR · **No calculator needed**

In a school, a student is chosen at random. The probability that the student is in grade 10, given that the student is in the robotics club, is $\dfrac{3}{8}$. The probability that the student is in the robotics club, given that the student is in grade 10, is $\dfrac{1}{4}$. There are 96 grade 10 students. How many robotics club members are NOT in grade 10?

Answer: 40

Grade 10 robotics members $= \frac{1}{4} \cdot 96 = 24$. These are $\frac{3}{8}$ of the club, so the club has $\frac{24}{3/8} = 64$ members. Not in grade 10: $64 - 24 = 40$.

Trap: Students use the same denominator for both conditional probabilities.

---

78. PS-169 · Hard · SPR · **No calculator needed**

In grade 10, 12 students are in a club and 28 are not. In grade 11, k students are in the club and 42 are not. Being in the club is independent of grade. What is the value of k?

Answer: 18

Independence means the proportion in the club is the same in each grade: $\dfrac{12}{40} = \dfrac{k}{k+42}$. $0.3k + 12.6 = k \Rightarrow k = 18$.

Trap: Students set $k = 12$ (equal counts) or compare with the wrong total.

---

79. **PS-170** · Hard · Multiple choice · **No calculator needed**

In grade 10, 18 students are in the robotics club and 42 are not. In grade 11, 30 students are in the robotics club and 60 are not. A student is chosen at random from the students who are in grade 10 OR in the robotics club. What is the probability that the student is in grade 11?

A) $\frac{1}{3}$

B) $\frac{1}{5}$

C) $\frac{1}{2}$

D) $\frac{5}{8}$

Answer: (A) $\frac{1}{3}$

The group is all 60 grade 10 students plus the 30 grade 11 robotics members: 90 students. Grade 11 students in it: 30. Probability $= \frac{30}{90} = \frac{1}{3}$.

Trap: Students use only the robotics club ($\frac{30}{48}$) or all students ($\frac{30}{150}$) as the denominator.

---

80. **PS-171** · Hard · SPR · **No calculator needed**

A line of best fit for a data set is $\hat{y} = 2.5x + 40$. For the data point with $x = 24$, the predicted value of y is 20% less than the actual value. What is the residual for this data point?

Answer: 25

Predicted $= 2.5(24) + 40 = 100$. $100 = 0.8 \cdot$actual, so actual $= 125$. Residual $=$ actual $-$ predicted $= 25$.

Trap: Students take 20% of the predicted value (20) or get the sign of the residual wrong.

---

81. **PS-172** · Hard · SPR · **No calculator needed**

The model $\hat{y} = 4.8x + 62$ estimates a town's population y, in thousands, where x is the number of decades since 1980. According to the model, what is the estimated increase in the population, in people, per year?

Answer: 480

4.8 thousand per decade $= 4{,}800$ people per 10 years $= 480$ people per year.

Trap: Students report 4.8, 48 or $4{,}800$ by converting only one of the units.

---

82. **PS-173** · Medium · SPR · **No calculator needed**

The ratio of grade 9 students to grade 10 students in a club is $3 : 5$, and the ratio of grade 10 students to grade 11 students is $4 : 7$. The club has 201 students in grades 9, 10 and 11. How many more grade 11 students than grade 9 students are in the club?

Answer: 69

Make the grade 10 terms equal: $12 : 20$ and $20 : 35$, so $12 : 20 : 35$ (sum 67). $201 \div 67 = 3$: 36, 60 and 105. Difference $= 105 - 36 = 69$.

Trap: Students write $3 : 5 : 7$ (sum 15) and divide 201 by 15.

---

83. **PS-174** · Hard · SPR · **No calculator needed**

A line of best fit for a data set is $\hat{y} = 2.4x + 15$. A new data set is formed by adding 5 to every x-value and keeping each y-value. What is the y-intercept of the line of best fit for the new data set?

Answer: 3

Shifting every x right by 5 shifts the line right by 5: $\hat{y} = 2.4(x - 5) + 15 = 2.4x + 3$. Same slope, new intercept 3.

Trap: Students add 5 to the intercept (20) or change the slope.

---

84. **PS-175** · Hard · SPR · **No calculator needed**

Data set A has 10 integers: 4 are from 10 to 19 and 6 are from 20 to 29. Data set B has 10 integers: 3 are from 30 to 39 and 7 are from 40 to 49. What is the smallest possible value of (mean of B) $-$ (mean of A)?

Answer: 12

Make A as large as possible and B as small as possible. Greatest mean of $A = \dfrac{4(19)+6(29)}{10} = 25$. Least mean of $B = \dfrac{3(30)+7(40)}{10} = 37$. Smallest difference $= 12$.

Trap: Students use bin midpoints (difference 21) or think the means cannot be compared.